MIL-C-45490B(MU)
18 December 1972
SUPERSEDING
MIL-C-45490A (MU)
7 June 1963

MILITARY SPECIFICATION

CARTRIDGE, 90MM, BLANK, M394 LOADING, ASSEMBLING AND PACKING

1. SCOPE

1.1 This specification covers the loading, assembling, and packing for one type of cartridge designated as Cartridge, 90MM, Blank, M394 for 90MM Guns.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-223	-Powder, Black
MIL-A-2550	-Ammunition and Special Weapons, General Specification for

STANDARDS

MILITARY

MIL-STD-105	-Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	-Quality Assurance Terms and Definitions
MIL-STD-1168	-Lot Numbering of Ammunition
MIL-STD-1235	-Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

ARMY

7549210	-Cartridge, 90MM, Blank, M394, for Guns, M36, M41 and M54 Assembly.
7549249	-Box, Packing, Ammunition for Cartridge, 90MM, Blank, M394, Assembly, Details, Packing and Marking.

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7549250	-Container, Ammunition, Fiber, M125A1, for Cartridge, 90MM, Blank, M394 Assembly, Details, Packing and Marking
6518063	-Chamber (Profile and Alignment)
6518064	-Plug, Alignment
6518092	-Check, Wear limit

Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Material.-Material shall be in accordance with applicable drawings and specifications.

3.2 Blank cartridges.-The blank cartridge shall comply with all requirements specified on Drawing (dwg.) 7549210 and with all requirements specified in applicable specifications.

3.3 Primers.-The primers shall have the required number and location of flash holes as specified on the appropriate primer drawing.

3.4 Air pressure test of the primer and cartridge case assembly.-The cartridge case, with the primer assembled thereto, shall withstand a minimum (min.) air pressure of 5 pounds per square inch for a min. of 15 seconds without leaking, as evidenced by a drop in gauge pressure of 0.01 pound or more, when tested as specified in 4.3.2.

3.4.1 Air pressure test of cartridge assembly.-The completed cartridge shall withstand a minimum air pressure of 5 pounds per square inch for a minimum of 15 seconds without leaking (as evidenced by a drop in gauge pressure of 0.01 pound), when tested as specified in 4.3.2.

3.5 Proving Ground Test (see 4.3.3)

3.5.1 Functioning and Sound level.-The sound level of the blank ammunition, at ambient temperature, shall be not less than 117 decibels when tested as specified in 4.3.3.

3.4.2 Extraction.-The cartridge case shall freely extract from the gun after firing.

3.5.3 Fragments.-There shall be no fragments thrown more

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than 300 feet beyond the weapon muzzle.

3.6 Workmanship.-All parts shall be free from dirt, burrs, cracks, chips, grease, rust, corrosion and other foreign material. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agents.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 to define terms used herein. The provisions of MIL-A-2550 shall apply.

4.1.1 Submission of product.-At the time each completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all quality assurance provisions specified within this specification.
b. Number of units of product inspected.
c. Results obtained, by defect number and paragraph, for all inspections performed.
d. Drawing, specification number and date, together with an identification and date of changes.
e. Certificates of conformance on all material purchased by the contractor when such material is controlled by Government or commercial specifications referenced in any of the contractual documents.
f. Number of items in the lot.
g. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

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4.2 Inspection provisions

4.2.1 Lot formation.-The term "inspection lot" as used in this specification is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only the quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revision, same specifications and same specification revision and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to the process, specifications, or drawings not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168. Each lot shall contain:

a. Primer assemblies of one lot interfix number from one manufacturer.
b. Black powder of one lot interfix number from one manufacturer.
c. New or fired cartridge cases modified by one manufacturer, of the same type of metal, from one lot interfix number.
d. Closing cups from one manufacturer.
e. Closing wads from one manufacturer.

4.2.2 Examination.-The examinations listed in the following subparagraphs shall be performed on inspection lots (see 4.2.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for critical defects (and major defects, when so specified) shall be 100 percent. Sampling plans for major and minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL's designated herein to the groups of major and minor defects listed, the contractor may elect to apply AQL's to individual defects using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. When this method is elected, it shall be applied to all of the major and minor defects listed in the applicable

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subparagraph of this section, except where 100 percent inspeciton is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.2.4.

4.2.2.1 Cup, closing (see dwg. 7549212 covering a detail of dwg. 7549210).

Categories	Defects	Method Of Inspection	
Critical: None defined			
Major:	AQL 0.40 percent		
101.	Cup cracked, broken, punctured or incompletely formed	Visual	
Minor:	AQL 1.5 percent		
201.	Outside diameter	Gage	CD
202.	Hole missing	Visual	
203.	Evidence of poor workmanship (see 3.6)	Visual	

4.2.2.2 Wad, closing (see dwg. 7549211 covering a detail of dwg. 7549210).

Categories	Defects	Method Of Inspection	
Critical: None defined			
Major: None defined			
Minor:	AQL 1.0 percent		
201.	Outside diameter, min	Gage	CD
202.	Evidence of poor workmanship (see 3.6)	Visual	

4.2.2.3 Case, cartridge (empty) (see dwg. 8857215 covering a detail of dwg. 7549210).

Categories	Defects	Method Of Inspection	
Critical: None defined			
Major:	AQL 0.40 percent		
101.	Distance from bottom of primer couterbore to front of seating flange	Gage	CD
102.	Crack or split extending through side wall or bottom	Visual	

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Minor:	AQL 1.5 percent	
201.	Total length	Gage CD
202.	Incorrect protective finish (applicable to steel cases only)	Visual
203.	Evidence of poor workmanship (see 3.6)	Visual

4.2.2.4 Loaded primer, prior to assembly to cartridge case (see dwg. 8839453 covering a detail of dwg. 7549210)

Categories	Defects	Method Of Inspection
Critical:		
1.	Any flash hole missing (see 3.3)	Visual
Major:	None defined	
Minor:	None defined	

4.2.2.5 Assembly of primer and case (see dwg. 7549210)

Categories	Defects	Method Of Inspection
Critical:		
1.	Primer above flush	Visual-Straight-Edge
Major:	AQL 0.40 percent	
101.	Primer can be pushed out of cartridge case by applying hand pressure against approved object held against primer	Visual-Manual
Minor:	AQL 0.65 percent	
201.	Foreign matter in interior of cartridge case	Visual

4.2.2.6 Assembly after sealing closing wad, before inserting closing cup (see dwg. 7549210).

Categories	Defects	Method Of Inspection
Critical: None defined		
Major: None defined		
Minor:	AQL 0.65 percent	
201.	Powder exposed	Visual

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4.2.2.7 Assembly after inserting wad and closing cup (see dwg. 7549210).

Categories	Defects	Method Of Inspection
Critical: None defined		
Major:	AQL 0.40 percent	
101.	Seal not continuous or groove not filled with cement	Visual
102.	Air escape hole not filled with cement	Visual
Minor: None defined		

4.2.2.8 Assembly taken from unsealed fiber container (see dwg. 7549210)

Categories	Defects	Method Of Inspection
Critical: None defined		
Major:	AQL 0.65 percent	
101.	Chamber gage (profile and alignment failure)	Gage 6518063
102.	Assembly damaged	Visual
103.	Closing cup loose or damaged	Visual-Manual
Minor:	AQL 1.5 percent	
201.	Marking misleading or unidentifiable including incorrect color code	Visual
202.	Glue or asphalt on exterior of cartridge	Visual
203.	Evidence of poor workmanship (see 3.6)	Visual

4.2.2.9 Unsealed fiber container (see dwg. 7549250)

Categories	Defects	Method Of Inspection
Critical: None defined		
Major:	AQL 0.40 percent	
101.	Packing component missing	Visual
Minor:	AQL 1.0 percent	
201.	Cartridge improperly assembled in container	Visual
202.	Cartridge cannot be removed by hand	Manual

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4.2.2.10 Sealed fiber container (see dwg. 7549250)

Categories	Defects	Method Of Inspection
Critical:	None defined	
Major:	AQL 0.65 percent	
101.	Tape incomplete or inner layer not in contact with the surface of the container for at least 1/2 inch on each side of joint	Visual
102.	Metal end loose or distorted	Visual
103.	Container damaged through all impregnated asphalt layers as evidenced by removing each layer of suspected container	Visual
104.	DOD or FSC symbol missing	Visual
Minor:	AQL 2.5 percent	
201.	Length of tear tab, min	Visual
202.	Color of tape incorrect	Visual
203.	Marking misleading or unidentifiable	Visual
204.	Cut, scuffs, or gouges in outer layer	Visual
205.	Gap between body and cover of container exceeding 1/8 inch	Visual-Scale
206.	Contents loose upon shaking	Manual

4.2.2.11 Sealed wood box (see dwg. 7549249)

Categories	Defects	Method Of Inspection
Critical:	None defined	
Major:	AQL 0.65 percent	
101.	Contents of box exposed	Visual
102.	DOD or FSC marking missing or incorrect	Visual
103.	Board broken or split	Visual
104.	Strapping, missing or loose	Visual-Manual
Minor:	AQL 1.5 percent	
201.	Marking misleading or unidentifiable	Visual
202.	Strapping improperly assembled	Visual-Manual
203.	Car seal missing, unsealed or incorrectly positioned	Visual-Manual

4.2.3 Testing

4.2.3.1 Moisture content of black powder (see dwg. 7549210)

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Major defect.-A sample of one cartridge, prior to sealing, shall be randomly selected from each hours production. The black powder shall be tested as specified in 4.3.1 and any failure to comply with the applicable requirement shall be cause for rejection of the eight hours production represented by the sample.

4.2.3.2 Air pressure test of primer and cartridge case assembly (see 3.4) - Major defect .-The acceptance and rejection criteria for this test shall be in accordance with Standard MIL-STD-105, using code letter J and AQL of 0.65. The test shall be performed as specified in 4.3.2.

4.2.3.2.1 Air Pressure Test of Cartridge Assembly.-(See 3.4.1)-Major defect-The test shall be performed 100 percent. Any cartridges failing to comply with the specified requirement shall be rejected and removed from the lot. This test shall be performed as specified in 4.3.4.

4.2.3.3 Proving Ground

4.2.3.3.1 Functioning and Sound Level (see 3.5.1) Major defects.-A sample of 35 cartridges shall be randomly selected from each lot for this test. The test shall be performed as specified in 4.3.3. If two or more cartridges fail to comply with the specified requirements, the lot shall be rejected. If only one cartridge fails to comply with the specified requirements, a second sample of 70 cartridges shall be randomly selected and tested. If, in the combined first and second samples, a total of two (2) or more sample cartridges fail to comply with the specified requirements, the lot shall be rejected.

4.2.3.3.2 Extraction (see 3.5.2) and fragments (see 3.5.3) Major defects.-The samples used in the functioning test (see 4.2.3.3.1) shall be used to determine compliance with requirements for extraction and fragments. If any cartridge fails to extract freely the lot shall be rejected (applies to both the first and second sample). If more than one cartridge fails the fragment requirement in the first sample, or in the event a second sample is required, if more than three cartridges fail in the combined test the lot shall be rejected.

4.2.3.3 Check test for possible deterioration of primers (see applicable primer specification)-Major defect.-If the total elapsed time between the original acceptance of any primer lot and the assembly of that lot into complete cartridges exceeds two years or if the primers have been subjected to adverse conditions, however brief, at any time since previous tests, the primer lot shall be subjected to and must satisfactorily pass, the check test specified in the applicable primer loading

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specification immediately before the primer lot is assembled into cartridges. This test shall be performed by the contractor on primers selected by the Government inspector at the facility assembling the primers into the cartridges (see 6.2).

4.2.4 The inspection equipment required to perform the examinations and tests prescribed in this section is identified under the "Methods Of Inspection" headings in 4.2 and 4.3 (see 6.3).

4.2.4.1 Government rights to documentation.-Inspection equipment drawings and lists provided and revised in accordance with the requirements of the specification may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

4.2.4.2 Government use of contractor's inspection equipment.-The contractor shall as required make available to the Government all inspection equipment necessary for determining conformance with contract requirements. Operation of the equipment and verification of its accuracy shall be supplied by the contractor, as required by the Government in performance of its examination or test.

4.3 Test methods and procedures

4.3.1 Moisture content of black powder.-The moisture content of the black powder shall be determined in accordance with the procedure specified in Specification MIL-P-223.

4.3.2 Air pressure test of the primer and cartridge case assembly. The primer and cartridge case assembly shall be subjected to an air pressure test using testing equipment in accordance with 4.2.4. Observation shall be made for compliance with the applicable requirements. CD

4.3.3 Functioning.-This test shall be performed by the Government at a proving ground or such other place as designated by the procuring activity. The cartridge shall then be fired at ambient temperature. Sound level shall be recorded using a sound meter located 100 feet from the gun muzzle on a line perpendicular to the axis of the tube. Testing equipment shall be in accordance with 4.2.4. Observation shall be made for compliance with the applicable requirements.

4.3.4 Air pressure test of complete cartridge assembly.-The

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complete cartridge assembly shall be subjected to an air pressure test using test equipment in accordance with 4.2.4. Observation shall be made for compliance with the applicable requirements. CD

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging

5.1.1 Level A.-The blank cartridge shall be preserved and packaged in accordance with dwg. 7549250.

5.2 Packing

5.2.1 Level A.-The box shall be packed in accordance with dwg. 7549249.

5.3 Marking.-Marking shall be in accordance with dwgs. 7549250 and 7549249 as applicable.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.
b. Data cards.-Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.

6.2 Cost of check test.-The contracting officer will arrange for the contractor to be reimbursed for the expense incurred in performance of the check test. The test shall be conducted at Government expense without cost to the contractor who loaded the primers or to the contractor assembling the primers in the cartridges and shall not constitute a basis of rejection against either contractor except where deterioration has occurred as a direct result of carelessness in handling, storage, etc. permitted while the primer lots were under the jurisdiction of either contractor.

6.3 Inspection Equipment Designs.-Inspection equipment designs are of two types-Government Special Inspection Equipment (SIE) designs and contractor designs. SIE designs are designated by drawing numbers or under the "Method of Inspection" headings in Section 4. Design responsibility for all other inspection equipment is assigned to the contractor. Equipment to be designed by the contractor is designated "CD". The contractor need not furnish any design when a complete Government SIE design is provided. Unless otherwise specified, however, the

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contractor may submit alternate or modified contractor designs of SIE in accordance with 6.3.2 and 6.3.3 should he elect to do so.

6.3.1 SIE designs.-SIE designs may consist of any of the following:

a. Detailed drawings which completely depict all information necessary for the fabrication and use of the item of inspection equipment.

b. A source control drawing or a specification control drawing as defined in MIL-STD-100.

c. An envelope drawing, as defined in MIL-STD-100, which establishes the criteria which a detail design shall meet. When envelope drawings are specified, the contractor shall prepare designs which comply with the criteria therein.

6.3.2 Contractor designs.-Contractor designs are required for all inspection equipment for which SIE designs are not specified and may include commercial equipment which the contractor proposes to use. (Commercial equipment is defined as unmodified equipment which is cataloged and available for purchase by the general public). Contractor designs shall include appropriate operating instructions, calibration procedures and maintenance procedures. Commercial equipment shall be fully described by catalog listings or other means which provide sufficient information to permit identification and evaluation by the Government and may include illustrations and engineering data. Designs shall be prepared for any special fixture(s) required to be used with commercial equipment, or with SIE designs if not otherwise covered thereby (see 6.3.1c). Design shall be of the category and form (per MIL-D-1000) specified in the Contracto Data Requirements Lists (DD Form 1423). The contractor is referred to MIL-HDBK-204, "Inspection Equipment Design" for guidance. The specification number and other specific identifying information from Section 4 of this specification shall be referenced on each contractor design together with the component or assembly drawing number and revision letter to which the specific design applies.

6.3.3 Submission of designs for approval.-Contractor designs shall be approved by the Government prior to fabricating or procuring the equipment. Designs shall be submitted for approval in accordance with the stipulations, time frame and distribution specified in the Contract Data Requirements List (DD Form 1423) or in the contract. Partial submission of inspection equipment designs is permissible and encouraged. However the completion date for design review will be based on the

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date of the final submission of designs. Picatinny Arsenal review will normally be accomplished within one month after receipt.

6.4 Submission of test data.-In addition to normal distribution of records, when the cartridge is procured by the Department of the Army, one copy of all ballistic test data will be forwarded to the following:

Commanding Officer
Picatinny Arsenal
Dover, New Jersey
ATTN: SMUPA-QA-A and SMUPA-AD-E-P

Custodian:
ARMY-MU

Preparing Activity
ARMY-MU

Project Number: 1315-A-008

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